

# Trainer Debrief — Lab 19

## ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)

**Case study:** Meridian Medical Devices — seal weld burst pressure

**Objective:** Compare multiple groups using ANOVA and apply non-parametric alternatives (A3, K2).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

### Expected analysis

lab-19-anova-interactions-nonparametric.xlsx

OneWay: means are A 224.8, B 226.2, C 204.3, D 221.8.  $F \sim 14$ ,  $p < 0.001$  — at least one supplier differs. Tukey shows Supplier C is significantly lower than A, B and D (which do not differ from each other). Act on Supplier C, not on 'suppliers' in general. TwoWay: Plant  $p < 0.001$ , Shift  $p \sim 0.03$ , and the Plant x Shift INTERACTION  $p < 0.001$ . Because the interaction is significant you must NOT interpret the main effects alone — Suzhou shift C is the specific bad cell, matching Labs 15 and 17. NonParametric: the data are strongly right-skewed, so ANOVA is invalid; Kruskal-Wallis  $H \sim 24.6$ ,  $p < 0.001$  — Line 3 cycle times are significantly higher.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

### What good looks like

Your ANOVA assumptions are tested and documented, significant results carry a post-hoc analysis, you have interpreted an interaction plot, and at least one finding is confirmed non-parametrically.

### Where this sits in the capstone

BLACK BELT ONLY. Comparing three or more groups with repeated t-tests inflates the false-positive rate badly — ANOVA solves this. Two-way ANOVA then reveals interaction effects that one-factor analysis cannot see. And when normality fails, non-parametric tests are the correct answer rather than proceeding regardless. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

### The running process — figures that must reconcile

| Checkpoint           | Value                                         | Established in |
|----------------------|-----------------------------------------------|----------------|
| Baseline defect rate | 2.9%                                          | Lab 3          |
| Measurement system   | FAILS Gage R&R (~40% study variation)         | Lab 11         |
| Baseline capability  | Pp 0.62 / Ppk 0.53                            | Lab 14         |
| Proven Xs            | clamp pressure, dwell time, electrode tip age | Labs 18-21     |
| DOE optimum          | clamp 47.4 bar, dwell 2.76 s                  | Labs 23-25     |
| Improved capability  | Ppk 1.40                                      | Lab 27         |

|                     |                                    |            |
|---------------------|------------------------------------|------------|
| Annual hard benefit | SGD 487,000 vs SGD 812,000<br>COPQ | Labs 1, 30 |
|---------------------|------------------------------------|------------|

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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